MAHATMAGANDHIUNIVERSITY,KOTTAYAM MGU
- UGP (HONOURS)
THIRDSEMESTEREXAMINATION
(2024ADMISSIONONWARDS)

MG3DSCECT203 INTEGRATEDELECTRONICS

Duration:1.5Hours **MaximumMarks:50**

Remember(K),Understand(U),Apply(A),Analyze(An),Evaluate(E)

PartA
MultipleChoiceQuestions.Answer**All**Questions.
Eachquestioncarries1mark

1. TheoperatingpointofaBJTisalsoknownasthe **K1**
   a) Cut-offpoint b)Saturationpoint
   c)Quiescentpoint d)Breakdownpoint
2. Negativefeedbackinanamplifiergenerallyleadsto: **U1**
   a) Increasedgain b)Increaseddistortion
   c)Increasedbandwidthd)Increasednoise
3. AnEmitterfollowercircuitisalsoknownasa: **K1**
   a) CommonEmitteramplifier
   b) CommonBaseamplifier
   c) CommonCollectoramplifier
   d) Darlingtonpair
4. Ashiftregisterusedforparallel-in,serial-outdata transfer **U2**
   requireshowmanyclockpulsestooutput N bits?
   a) 1 b)N c)N-1 d)2N
5. WhichofthefollowingisanidealOp-Amp parameter? **K3**
   a) Finiteinputimpedance
   b) Infiniteoutputimpedance
   c) Zeroinputoffsetvoltage
   d) Limitedslew rate
6. Inthefrequencyresponseofalow-passfilter,thegainstarts **U3**
   toroll offat arateofapproximately:
   a)+20dB/decadeb)20-dB/decadec)0dB/decaded)+40dB/decade
7. Ifa3-bitasynchronousupcounterisinitiallyat000,whatwillbeits **A2**
   stateafter5clock pulses?
   a) 010 b)101 c)011 d)110
8. TheNE555timerICiscommonlyusedto implement: **U3**
   a) Voltage-controlled oscillatorb)Astablemultivibrator
   b) Audioamplifier d)Powersupply
9. IfanOp-AmphasaverylowSlewRate,whichtypeofoutputsignal willit **U3**
   struggletoreproduceaccurately?

a) DCvoltage b)Low-frequencysinewave

c)High-frequencysquarewave d)Constantvoltage

10. A**JohnsonCounter**with'N'flip-flopshashowmanyuniqueoutputstates? **U2**

a) N b)2N c)2^N d)2N-1

10*1 = 10

PartB

ShortAnswerQuestions.Answer**4**Questions.
EachQuestioncarries5marks

11. Explainthepurposeandworkingofavoltagedividerbiasingcircuitfora BJT. **U1**
Discusshowithelpsstabilizetheoperatingpoint(Q-point)againstvariationsintransistor parameters.
12. StateandexplaintheBarkhausenCriteriaforsustainedoscillationsinanoscillator **K1**
circuit.
13. IllustratetheoperationofaRingCounterwithasuitablediagram andstateits **U 2**
application**.**
14. Listandexplainanythreeimportantparametersofan Op-Amp. **U 3**
15. Differentiatebetweenopen-loopandclosed-loopOp-Ampconfigurations,highlightingA**n3**
theirkeycharacteristics.
16. For an ideal Op-Amp, design an inverting amplifier circuit to achieve a voltage gain of -10
Assumeasuitableinputresistorvalueandcalculatetherequiredfeedbackresistorvalue.**A3**

5*4 = 20

PartC

EssayQuestions.Answer**any2**Questions Each
question carries 10 marks

17. DescribethecompleteoperationofasingleRCcoupledamplifierusingBJT **U1**
withvoltagedividerbiasing,includingacircuitdiagramandthefunctionofeachcomponent.
18. Designa3-bitsynchronousUP/DOWNcounterusingJKflip-flops.Drawthecomplete**An2**
logicdiagramandexplainitsoperationwithdetailedtimingdiagramsforbothup-counting and down-counting sequences.
19. AnalyzetheoperationofanOp-AmpasaSchmittTrigger.Drawitscircuit diagram **An3**
andexplaintheconceptofhysteresiswithitsinput-outputtransfercharacteristics.
20. DiscusstheconceptofanOp-AmpactingasaComparator.Explainthedifference **A 3**
betweenabasiccomparatorandaZero-CrossingDetector.Drawcircuitdiagramsforboth and illustrate their input-output waveforms.

10*2 = 20